is to make use of the brute power of computers to simulate the resulting motion.

This example highlights the importance, when using a perturbative approach, of determining whether the supposedly ballpark estimate really is in the ballpark, and if it is, which and how many of the finer details must be included in order to achieve a desired level of accuracy. As we now discuss, these issues are particularly crucial for applying perturbative tools to physical processes in the microworld.

A Perturbative Approach to String Theory

Physical processes in string theory are built up from the basic interactions between vibrating strings. As we discussed toward the end of Chapter 6,* these interactions involve the splitting apart and joining together of string loops, such as in Figure 6.7, which we reproduce in Figure 12.3 for convenience. String theorists have shown how a precise mathematical formula can be associated with the schematic portrayal of Figure 12.3—a formula that expresses the influence that each incoming string has on the resulting motion of the other. (The details of the formula differ among the five string theories, but for the time being we will ignore such subtle features.) If it weren't for quantum mechanics, this formula would be the end of the story of how the strings interact. But the microscopic frenzy dictated by the uncertainty principle implies that string/antistring pairs (two strings executing opposite vibrational patterns) can momentarily erupt into existence, borrowing energy from the universe, so long as they annihilate one another with sufficient haste, thereby repaying the energy loan. Such pairs of strings, born of the quantum frenzy but which live on borrowed energy and hence must shortly recombine into a single loop, are known as virtual string pairs. And even though it is only momentary, the transient presence of these additional virtual string pairs affects the detailed properties of the interaction.

*Those readers who skipped over the "More Precise Answer" section of Chapter 6 might find it helpful to skim the beginning part of that section.

This is schematically depicted in Figure 12.4. The two initial strings slam together at the point marked (a), where they merge together into a single loop. This loop travels a bit, but at (b) frenzied quantum fluctuations result in the creation of a virtual string pair that